[pending: working title — the block-ridge arc]

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Abstract

[pending: abstract — written last, once the arc is frozen]

1 Introduction

[decision pending]

2 Data, Preparation, and Descriptive Analysis

2.1 Data

The raw data are 30-minute bars for the S&P 500 complex, gridded to a regular 30-minute frequency. The panel's extent is not set by a chosen date: **a bar exists in the panel if and only if the realized-variance source printed on it** — the same observed-print fact the availability indicators of Section 2.2 are built from, applied as the grid's own existence criterion. Under this rule the panel begins at the source's first print, in January 1998, and ends at its last, in April 2024. No weekend rule, holiday calendar, or session table is supplied anywhere: weekends, holidays, early closes, and sessions that do not yet exist are all absent from the panel for the same reason — the source did not print.

The panel's session structure therefore evolves as the market's did. The source prints around the clock from its first day — the panel's first bar is an overnight bar — but the early sessions are sparser: roughly 7,500 bars per year in 1998 (median 32 bars per trading day before 2004), growing through roughly 10,600 bars in 2003 to a stable $\approx$ 12,400 bars per year from 2005 onward, when the near-24-hour cycle (Sunday 18:30 through Friday 20:00, 48 bars per full trading day, roughly 34 of them outside regular hours) is fully live. A time-of-day slot that thickens or enters mid-sample behaves exactly like a feed going live: the rolling per-slot estimators of Section 2.2 have whatever history the slot has and accumulate it causally. Any statement about intraday structure in this panel refers to the full cycle as it stood at the time, not to the 09:30–16:00 window alone.

The existence rule replaced an earlier calendar-only filter (weekend trim plus forward fill) after that filter produced a measurable incident: bars following an early close existed on the calendar grid as forward-filled ghosts, and on one such bar — the only one in more than twenty-five thousand — two feeds' availability indicators disagreed for a single bar, with consequences documented in Section 4. On the overlapping 2005–2024 span the rule removes 4,925 such dead-session bars ($\approx 2\%$ of the calendar grid). **[pending: v3 panel counts from the harvest: grid rows, panel rows after burn-in, and scored evaluation rows after the training window and the calibration burn-in of Section 3]**

The forecasting target is realized variance. For each 30-minute bar it is computed as the sum of squared 5-minute log-returns within the bar:

$$RV_{d,j} = \sum_{i=1}^{M} r_{d,j,i}^2, \qquad M = 6, \tag{1}$$

where $r_{d,j,i}$ is the i-th 5-minute log-return inside the j-th 30-minute bar of day d. Bars are indexed sequentially as RV_t. The forecast horizon is one bar ahead: the model at bar t predicts RV_{t+1}. Throughout, "realized variance" denotes the target RV_t, and "volatility" denotes its square root; models are estimated on the square-root scale (Section 2.2). The construction in Equation (1) is the standard realized-measure definition. The heterogeneous autoregressive model of Corsi [2009] uses the trading day as its base unit; here both the target and the lag ladder are defined at 30-minute resolution.

The evaluation panel. Feature construction consumes a fixed burn-in of 3,125 bars at the start of the sample (January to June 1998), a constant of the panel construction that exceeds the longest feature lookback. This reduces the grid's 303,442 print-bars to a usable panel of 300,317 bars. The walk-forward protocol consumes a further 24,000 bars as the initial training window, and the calibration burn-in of Section 3 consumes the first evaluation window. Every arm reported in this paper is scored on the identical remaining rows; the paired bar-for-bar comparisons used throughout require this shared row index. The scored evaluation panel contains $n = 273{,}554$ bars.

Exogenous information. Beyond the target's own history, the design uses 41 exogenous channels. The channels are organised into thematic buckets that partition the columns exactly $(7 + 12 + 6 + 6 + 3 + 3 + 4 = 41)$:

- **Moments** (7): own-asset higher-order return moments: bipower variation, realized semi-variance, autocovariance, absolute, cubic, and quartic returns.
- **Liquidity** (12): trading volume, turnover (total, buy-initiated, sell-initiated), effective and quoted spreads, number of trades.
- **Market, equal-weighted** (6) and **value-weighted** (6): cross-sectional stock-factor aggregates of return moments, bipower variation, semi-variance, and absolute returns.
- **Sentiment** (3): StockTwits attention, sentiment score, and sentiment count.
- **Implied volatility** (3): VIX, VVIX, VIX3M.
- **Volatility demand** (4): options order-flow indicators (SPX open/close, all options open/close, open only).

The buckets differ in column count; liquidity contributes twelve columns and sentiment three. A per-bucket comparison therefore mixes information content with dimensionality unless it is read per feature. Availability also differs across buckets: the cross-sectional aggregates are observed on 138,357 bars, the options order-flow bucket on 54,062 bars, and the VIX family does not exist for the earliest part of the sample. Row deletion would score each bucket on a different sub-sample. Section 2.2 describes the impute-and-indicate construction used instead; it scores every bucket on the same rows.

[pending: feature accounting: a per-bucket table giving channels, the lag set applied to each, derived columns, and the reconciliation of the three circulating counts (41 channels, 359-column exogenous block, 529-column design) — the stated expansion rule of six rolling means per channel gives $41 \times 6 = 246$ and does not reconcile them; no per-column claim should be read as final until this table exists]

2.2 Preparation

This subsection specifies the preparation pipeline. The pipeline is presented in full because, measured on this panel, the representation of missing and stale data has a larger effect on QLIKE than any model-class, basis, or penalty decision reported in this paper; the magnitudes are given under "Rolling robust scaling" below. At any given bar, 36% of the exogenous design consists of forward-filled values rather than new measurements; on the sparsest channels the figure is 78%. Without availability information, an estimator cannot distinguish a filled value from a new measurement.

Overnight fills and availability indicators. The 41 exogenous channels are not all published on the near-24-hour grid. Most are US cash- or extended-session equity quantities; each prints only inside its own session. Averaged across channels, 0.638 of panel rows carry a print from the raw feed and the remaining 0.362 do not. Each channel is processed in the following order:

1. Compute a binary availability indicator from the raw feed: the indicator is 1 on bars where the feed printed and 0 otherwise.
2. Forward-fill the channel within its availability window.
3. Where the channel is structurally absent (no prior print exists), substitute a fixed in-distribution neutral value.

Every value column is accompanied by its indicator column. The indicator is computed before the fill. An earlier build of this pipeline computed the indicator after the fill, so every forward-filled bar was labelled observed. Under that build, the volatility-demand feed stopped publishing on 2023-08-31, the unlimited forward fill carried its final August print across the following eight months (8,453 bars, 3.44% of the sample), and the indicator read ≈ 1 throughout. Three further feeds (`vix`, `vix3m`, `vvix`) stopped publishing the same way on 2024-02-12. Every built cache is now gated on one check: for each channel, the mean of the constructed availability indicator must equal the fraction of rows on which the raw feed printed. The two quantities are estimates of the same number, so any gap indicates a construction error of this kind. On the uncorrected panel the check fails for 31 of 41 channels. The check requires no model, no held-out data, and no threshold tuning. We recommend it for any mixed-frequency panel in which exogenous series are filled.

The indicator also enters estimation as a regressor. Its linear coefficient absorbs any level shift between the pre- and post-availability eras, and the value column contributes only where the feed exists. Every bucket, including implied volatility and volatility demand, is therefore scored on the same 218,934-row index. The exact paired significance tests of the later sections require this shared index.

Stage 1: diurnal adjustment. Intraday volatility has a time-of-day profile. Let $s(t) \in \{1, \dots, 48\}$ index the time-of-day slot of bar t, and let $\mathcal{W}_t = \{j : j < t,\ s(j) = s(t)\}$ be the set of the most recent $W = 20$ observations sharing that slot, with at least 5 required before the baseline is valid. For non-negative series, the baseline is the same-slot rolling mean and the adjustment divides by it:

$$\bar{RV}_{t|s} = \frac{1}{|\mathcal{W}_t|} \sum_{j \in \mathcal{W}_t} RV_j, \qquad \widetilde{RV}_t = \frac{RV_t}{\bar{RV}_{t|s}}, \tag{2}$$

Signed series (e.g. autocovariance) are divided by the same-slot rolling standard deviation instead. The window condition is $j < t$, not $j \leq t$, so the baseline at bar t uses no information from bar t; the adjustment is non-anticipative. Warm-up bars pass through with baseline 1. They are consumed by the burn-in and never enter the evaluation panel.

Stage 2: the winsorized square-root target. After the diurnal adjustment, each variable class receives a fixed variance-stabilising map. The map is chosen in advance from the class's support and tail behaviour; no transform parameters are fitted. The maps are: $x \mapsto \sqrt{x}$ for squared-return and volatility-like measures, including RV itself, bipower variation, turnover, and effective spread; $x \mapsto \operatorname{sign}(x)\sqrt{|x|}$ for autocovariance; $x \mapsto x^{1/3}$ and $x \mapsto x^{1/4}$ for third and fourth moments; and $x \mapsto \log x$ for the implied-volatility family. Models are estimated on the adjusted $\sqrt{RV}$ scale. After transformation, each series is clipped to rolling 1st and 99th percentile bounds estimated on the trailing $W_{\text{clip}} = 240$ observations (about five trading days) ending at $t-1$. Winsorization is not inverted at evaluation time; only the diurnal division and the square root are inverted. Because $\mathbb{E}[\sqrt{X}]^2 \neq \mathbb{E}[X]$, squaring a forecast of $\sqrt{RV}$ gives a downward-biased forecast of RV. We therefore apply Duan's smearing correction [Duan, 1983]: add the in-sample residual variance to the squared forecast, then multiply by the stored diurnal baseline to undo the division of Stage 1:

$$\widehat{RV}_t^{\text{raw}} = \left(\hat{f}_t^{\,2} + \hat{\sigma}_\varepsilon^2\right) \cdot \bar{RV}_{t|s}^{\text{baseline}}, \qquad \hat{\sigma}_\varepsilon^2 = \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2. \tag{3}$$

HAR features. The target's own history enters through trailing rolling means of the adjusted series on a base-2 geometric ladder,

$$\mathcal{L} = \{1,\ 2,\ 4,\ \dots,\ 2^{k_{\max}}\}, \qquad \overline{RV}_t^{(k)} = \frac{1}{k} \sum_{i=0}^{k-1} \widetilde{RV}_{t-1-i}, \tag{4}$$

The rungs span roughly 30 minutes, 2.5 hours, half a trading day, 2.6 days, 13 days, and 65 trading days. The sum starts at $t-1$, so the feature at bar t contains no contemporaneous information. The ladder adapts the daily/weekly/monthly averages of Corsi [2009] to six intraday horizons. The same ladder is applied to each included exogenous channel, contributing $|\mathcal{L}| = 6$ rolling-mean columns per channel.

Rolling robust scaling. Input features to the linear estimators are standardised by subtracting a rolling median and dividing by a rolling interquartile range. Both statistics are estimated from the current training buffer and updated at every step. Two rules govern the estimation.

First, the median and IQR are estimated from observed rows only, i.e. rows on which the availability indicator is 1. If the statistics are instead estimated over forward-filled values, a series whose feed has stopped contributes identical repeated values, its rolling IQR shrinks toward zero, and the standardising division divides by a value near zero. On the uncorrected panel, the stale volatility-demand columns reached 2,260 rolling IQRs by this mechanism; the raw series did not move, and the denominator shrank.

Second, a scale is used only after 512 observed values have accumulated, and the IQR is floored at 1% of the largest scale the channel has shown. This rule exists because masked estimation fails at the start of a feed's life: with few observed rows, a low-dispersion opening window would otherwise set the scale for all subsequent bars.

The three elements — indicators computed before the fill, masked estimation, and neutral substitution — are applied together. Each element applied alone does not remove the failure. Applied together, they reduce the worst channel's maximum standardised value from 2,260.4 to 96.2 rolling IQRs. On the October 2023 window affected by the stale feed, the uncorrected panel scored QLIKE 6.37170 and the repaired panel scored QLIKE 0.12391. Every modelling effect reported in this paper lies between 0.0005 and 0.004 QLIKE. This difference in magnitude is the reason the preparation pipeline is presented before the modelling sections.

All stages are strictly causal: every quantity dated t uses only information available strictly before t.

2.3 Descriptive Analysis

[decision pending]

2.4 The OLS–HAR Benchmark

Every result in the remainder of this paper is referenced to a single fixed comparator: ordinary least squares on the six HAR features of Equation (4) plus calendar dummies, with no exogenous variables, refit at every bar. The forecasting protocol is:

1. Walk-forward with a rolling origin. The training window is the most recent $T_{\text{train}} =$ 24,000 bars (approximately 500 trading days at 48 bars per day). The window slides forward by one bar per prediction.

2. The model is refit at every bar. The refit is exact and computationally feasible because the sliding-window normal equations admit rank-1 up-dating (add the entering bar) and down-dating (remove the exiting bar).

3. The benchmark is read from the ridge path at $\alpha = 0$; it is therefore exact OLS.

4. Each forecast is mapped back to raw variance through Equation (3).

5. Forecasts are scored by QLIKE [Patton, 2011],

$$L_{\text{QLIKE}} = \frac{1}{N} \sum_{t=1}^{N} \left[r_t - \log r_t - 1 \right], \qquad r_t = \frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}}, \tag{5}$$

computed on the full 218,934-bar evaluation panel. QLIKE is used for three reasons: it is scale-invariant in the level of volatility; it is strictly consistent for the conditional variance; and it penalises under-prediction more heavily than over-prediction, which matches the direction in which forecast errors are more costly for this target. QLIKE values are not comparable across preprocessing regimes: on the diurnally adjusted panel of the reference study, the same class of models scores several tenths of a QLIKE unit lower than on a raw, unadjusted panel. Every comparison in this paper is therefore made within one preprocessing regime, and every table is labelled with the panel it was computed on.

[pending: the benchmark table: per-bar OLS-on-HAR QLIKE and secondary metrics (out-of-sample R^2, Mincer–Zarnowitz α/β, MSE, MAE) from the unification campaign's `a0` arm on the frozen base-2 panel — the single benchmark every comparison in this paper is measured against]

Differences in QLIKE at the fourth decimal place require a significance assessment. Every comparison against this benchmark is therefore accompanied by a Diebold–Mariano test [Diebold and Mariano, 1995] on the per-bar loss differential. The test procedure is:

1. Compute the per-bar QLIKE loss for each of the two arms on the shared row index, and take the per-bar difference.

2. Estimate the long-run variance of the loss differential with a Newey–West estimator.

3. Apply the small-sample correction of Harvey et al. [1997] to the test statistic.

Where several arms are compared simultaneously, we report the Model Confidence Set of Hansen et al. [2011]. Every arm is scored on the identical row index, so all of these tests are paired bar-for-bar comparisons.

Hyperparameter discipline. No quantity anywhere in this paper is selected using evaluation loss. Every hyperparameter that is tuned is tuned *causally*: the penalty (or penalty vector) is re-selected periodically from a grid fixed in advance, by minimizing validation error on a forward split of the current training window — fit block, embargo, validation tail — so that no forecast's configuration uses information from that forecast's future. Every hyperparameter that is not tuned is a fixed constant, set before the campaign was scored and reported with its arm: the fixed-penalty arms exist precisely as controls for what tuning contributes, and the structural constants of the design (training-window length, re-selection cadence, winsorization quantiles, block widths, estimation-window lengths) are conventions of the panel and estimator construction, disclosed where they are defined and never optimized against any loss.

3 A Calibrated Second-Moment Correction

3.1 Why the Back-Transform Needs a Second Moment

Every model in this paper is fit on the preprocessed scale. Realized variance RV_t is diurnally adjusted and then passed through the square-root transform, so the regression target is $y_t = \sqrt{\widetilde{RV}_t}$, where $\widetilde{RV}_t$ denotes the diurnally adjusted variance at bar t. Evaluation is performed on the raw variance scale. This is because QLIKE — the primary loss — is a strictly consistent scoring rule for the conditional variance, not for any transform of it. A forecast issued in square-root space must therefore be mapped back to raw space before it is scored.

The naïve inverse squares the forecast and reverses the diurnal adjustment. Squaring does not commute with conditional expectation: by Jensen's inequality, $\mathbb{E}[\sqrt{X}]^2 \leq \mathbb{E}[X]$, with equality only in the degenerate case. Decompose the target as $y_t = \hat{y}_t + \varepsilon_t$, where $\hat{y}_t$ is the model's forecast of y_t and ε_t is a conditionally mean-zero innovation. Then

$$\mathbb{E}\left[y_t^2 \,\middle|\, \mathcal{F}\right] = \hat{y}_t^2 + \mathbb{E}\left[\varepsilon_t^2 \,\middle|\, \mathcal{F}\right], \tag{6}$$

where $\mathcal{F}$ is the conditioning information set. The conditional mean of the squared target exceeds the squared forecast by the conditional second moment of the square-root-scale innovation. A forecast that is well calibrated in $\sqrt{RV}$-space is, once squared, biased low in RV-space, and the size of the bias equals the residual variance of the model being evaluated.

Two facts about this bias determine the design. First, QLIKE penalizes under-prediction more heavily than over-prediction, and the bias of the uncorrected squared forecast is downward. Second, the bias equals each model's own residual variance, so it differs across models. The back-transform therefore includes an explicit second-moment term, and the estimator of that term is specified as part of the evaluation procedure.

3.2 The Calibrated Estimator

The correction is a two-component causal calibration: a mean map re-estimated at every window boundary on the window before it, and a second moment seeded there and adapted bar by bar. Fix the chunked walk-forward grid: the backtest is split into contiguous evaluation windows indexed $k = 0, 1, \ldots$, tiling the panel in row order. The inputs for window k are three per-bar arrays over its bars: the forecasts $\{\hat{y}_t\}$ on the transformed (square-root, diurnally adjusted) scale, the diurnal baseline factors $\{B_t\}$, where B_t is the factor that reverses the Stage-1 adjustment at bar t, and the raw realizations $\{RV_t^{\text{raw}}\}$. The evaluation-scale target is the *unwinsorized* transformed realization, reconstructed exactly from the persisted raw target,

$$y_t^{\text{raw}} = \sqrt{RV_t^{\text{raw}}/B_t}. \tag{7}$$

Mean calibration. The forecast is passed through one affine map per window,

$$m_t = \hat{a}_k + \hat{b}_k\,\hat{y}_t, \qquad (\hat{a}_k, \hat{b}_k) = \arg\min_{a,\,b} \sum_{s\,\in\,k-1} \big(y_s^{\mathrm{raw}} - a - b\,\hat{y}_s\big)^2, \tag{8}$$

the Mincer–Zarnowitz regression of the evaluation-scale target on the forecast, estimated by ordinary least squares over the valid bars of window $k-1$ and applied to every bar of window k. The map absorbs the level and slope miscalibration the fit-scale construction leaves behind, including the attenuation induced by training on the winsorized target. For a window whose predecessor is missing from the harvest — a mid-panel gap, which a complete harvest does not contain — the map falls back to the identity $(\hat{a}_k, \hat{b}_k) = (0, 1)$; every such window is flagged in the harvest accounting.

Conditional second moment. Let $e_s = y_s^{\mathrm{raw}} - m_s$ denote the residuals against the calibrated mean as applied at their own bar. The second-moment term is adaptive. Window k opens at the seed

$$\hat{\sigma}^2_{(k)} = \frac{1}{N_{k-1}} \sum_{s\,\in\,k-1} e_s^2, \tag{9}$$

the mean of window $k-1$'s squared residuals, and within the window the term tracks the model's own recent error level causally,

$$\hat{\sigma}^2_t = \lambda\,\hat{\sigma}^2_{t-1} + (1-\lambda)\,e^2_{t-1}, \qquad \lambda = 0.6, \tag{10}$$

so the variance applied at bar t uses residuals through bar $t-1$ only. The decay sits on a broad, flat optimum: every value of λ between 0.4 and 0.6 scores within 10^{-4} of the best on the benchmark arm, and $\lambda = 0.6$ is the consensus choice across the block ladder's rungs. Positivity is trivial — a convex combination of squares — and the estimator does not extrapolate: no value of window k's forecasts can move it. The scalar of Equation (9) held fixed across the window — the contract's first iteration — loses to the retrospective in-window Duan baseline on the benchmark (0.23353 against 0.23165) precisely because it cannot track the level; the adaptive form beats that baseline on every arm measured, while remaining strictly causal (benchmark 0.22519, paired ΔQLIKE -0.0065 at DM -5.7 against the Duan convention). For a window whose predecessor is missing from the harvest, the seed falls back to the in-window scalar mean of $(y_t^{\mathrm{raw}} - m_t)^2$; every such window is flagged in the harvest accounting.

Relation to the literature. Standard practice corrects the back-transform with a *static* scalar: the analytic log-normal factor $\exp(\hat{\sigma}^2_\varepsilon/2)$ of the log-RV literature, extended by Bekierman and Manner [2018] with the state variance of their state-space HAR, or Duan [1983]'s nonparametric mean of the exponentiated residuals. Whether the correction is worth applying at all is contested: Demetrescu et al. [2020] compare bias corrections for exponentially transformed forecasts, find none uniformly best, and recommend skipping the correction under high persistence. The origin of the second moment's predictability is the realized estimator's own measurement error [Asai et al., 2012], whose variance scales with the volatility level. To our knowledge no existing convention lets the correction itself track that level; Equation (10) is exactly the standard scalar correction with its level allowed to adapt causally, and the static scalar is recovered at $\lambda = 1$.

Calibration burn-in. The first evaluation window of each arm is the calibration burn-in, parallel to the feature construction's burn-in at the start of the panel: it estimates the two maps for its successor and is not scored — none of its bars enters any loss, any join, or any calibration diagnostic. Scoring begins at the second window. Every scored forecast is therefore computable at its own bar, with no exceptions.

The conditional-variance record. Two bar-conditional parameterizations of the second moment were implemented and rejected on measurement. An affine form, $\hat{\sigma}_t^2 = \max(\hat{c}_0 + \hat{c}_1 \hat{y}_t^2, 0)$, fit on the previous window's squared residuals, admits zero variance on quiet bars and lets the raw forecast collapse: under it, twenty bars carried 99% of one arm's pooled QLIKE. A log-linear form, $\hat{\sigma}_t^2 = \exp(\hat{d}_0 + \hat{d}_1 \hat{y}_t^2)\, \hat{S}$, with $\hat{S}$ the Duan [1983] smearing factor for the log-retransformation, is structurally positive but extrapolates exponentially at bars whose $\hat{y}_t^2$ lies outside the previous window's fitted range: the second moment reached 10^{44}, inflating every arm's loss and collapsing the Diebold–Mariano differentials toward zero. A bar-conditional second moment therefore requires a model with controlled extrapolation, which is the reserved learned-correction slot for this section; the contract uses the scalar of Equation (9).

A shrunk conditional second moment. The two rejected forms fail for the same reason: neither is anchored to the estimator that works. A third form removes that failure mode by construction. Let

$$z_t \;=\; \log\!\big((y_t^{\text{raw}} - m_t)^2\big),$$

and predict z_t by ridge regression on a standardized design, fit on the rolling window and refit at every bar. The penalty is selected causally on the same embargoed forward split used for every other hyperparameter in this paper, from a grid that includes $\alpha = \infty$; at that grid point the fitted value is the training-window mean of z and the assembled variance is *exactly* the window-level second moment of Equation (9) — the seed of the adaptive recursion (10). The window-level contract is therefore nested as the infinite-shrinkage limit, and conditioning is adopted only where the forward split prefers it. At finite α the variance is

$$\hat{v}_t \;=\; \exp(\hat{z}_t)\, c_k, \qquad c_k \;=\; \frac{\text{mean of } (y^{\text{raw}} - m)^2 \text{ on window } k-1}{\text{mean of } \exp(\hat{z}) \text{ on window } k-1},$$

so each window's variance *level* is pinned to the scalar estimator's level and conditioning can only redistribute variance across bars within a window. No floor, clip, or additive constant appears anywhere in the construction.

Measured against the benchmark on the shared panel, the conditional contract is the largest single effect in this study: applied to the benchmark's own residuals it improves QLIKE by -0.00453 at DM -12.0, and applied to the paper's best mean model by -0.00394 at -9.4. Both are an order of magnitude larger than any increment produced by changing the conditional mean. These conditional-variance runs were scored under the campaign's first contract — the window-level scalar of Equation (9) held fixed within each window; under the adaptive recursion of Equation (10) part of this gain is absorbed by construction, since the conditional model and the recursion trade on the same level channel, and re-scoring the sidecar under the new contract is reserved.

Which state carries the second moment. That result invites the reading that the exogenous panel predicts forecast uncertainty. A control arm rejects it. Conditioning the variance on the backbone alone — the autoregressive ladder and calendar dummies, with no exogenous column — improves the benchmark by -0.00535 at DM -15.3, and beats the exogenous-state version head to head (-0.00082 at -2.9). The predictable part of the squared forecast error is therefore carried by the volatility level itself, which is what the realized-variance estimator's own measurement error scales with, and not by the exogenous state. The conditional second moment is a large and robust effect about heteroskedasticity in the target; it is not evidence that the exogenous buckets carry distributional information beyond the level, and this paper does not claim that they do.

Back-transform. Both members of the scored pair are mapped to the raw scale as

$$\widehat{RV}_t^{\text{raw}} = \big(m_t^2 + \hat{\sigma}_t^2\big)B_t, \qquad RV_t^{\text{raw}} = \big(y_t^{\text{raw}}\big)^2 B_t, \tag{11}$$

where the right-hand identity is Equation (7) inverted: the realization is the persisted raw target itself, never reconstructed from the winsorized fit target. The traditional scalar estimator of Duan [1983] is the same form with the identity mean map, a window-constant scalar, and in-window fit-scale residuals: the contract differs from Duan smearing in exactly three ways — the calibrated mean, the adaptive recursion of Equation (10) in place of the window-constant scalar, and causal residuals against that mean on the evaluation scale in place of the in-window fit-scale residuals (which see the scored window's own errors).

The full scoring procedure for one evaluation window is:

1. *Inputs.* The per-bar arrays $\{\hat{y}_t\}$, $\{B_t\}$, and $\{RV_t^{\text{raw}}\}$ over windows $k-1$ and k; reconstruct y^{raw} by Equation (7).

2. *Calibration.* Estimate $(\hat{a}_k, \hat{b}_k)$ by Equation (8) and the seed $\hat{\sigma}^2_{(k)}$ by Equation (9) on window $k-1$'s valid bars, with the stated fallbacks.

3. *Back-transform.* Map every bar of window k to the raw scale with Equation (11), producing the per-bar pairs $\big(\widehat{RV}_t^{\text{raw}}, RV_t^{\text{raw}}\big)$.

4. *Loss.* Evaluate QLIKE on the raw-scale pairs,

$$\text{QLIKE} = \frac{1}{N}\sum_{t=1}^{N}\left(\frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}} - \log\frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}} - 1\right), \tag{12}$$

   where the mean is taken over the bars at which both members of the pair are positive; bars failing that condition are excluded from the mean.

Because $\hat{\sigma}_t^2$ is a convex combination of squares and $m_t^2 \geq 0$, the calibrated forecast $\widehat{RV}_t^{\text{raw}}$ is positive whenever B_t is and window $k-1$'s residuals are not identically zero; the positivity condition of the exclusion rule then binds only through the realization.

Three implementation choices define the estimator.

Window-level calibration, bar-level variance. The mean map $(\hat{a}_k, \hat{b}_k)$ is re-estimated at every window boundary from the window before it; the second moment opens at the previous window's scalar and then adapts bar by bar by Equation (10). On the raw scale the correction also varies through the baseline factor B_t. Relative to the squared calibrated forecast, the correction is largest at the bars where m_t^2 is smallest — the quiet bars — and largest in level immediately after the model's own recent errors have been large.

Causal, hence deployable. Every estimated quantity applied in window $k \geq 1$ is a function of data through window $k-1$ only, and windows are processed in strictly ascending order. The map is a quantity a deployed forecaster could have computed at t: the scored object is a real-time forecast, not a retrospectively corrected one. Scoring begins after the calibration burn-in, so there are no exceptions.

Residuals of the model itself, against the unwinsorized target. The residuals entering Equation (9) are the evaluated model's own out-of-sample forecast errors from the preceding window, measured against the unwinsorized evaluation-scale target of Equation (7) — not training-set residuals, not the residuals of any reference model, and not fit-scale residuals. The correction is therefore model-specific: an arm with larger residual variance receives a larger upward correction, so the correction is part of each arm's forecast, not a shared post-processing constant. With this choice the decomposition of Equation (6) holds for the evaluation target itself, tails included. One bias remains and is stated plainly: the *coefficients* of each arm are estimated on the winsorized fit target, so the forecast $\hat{y}_t$ itself carries the fit-scale clipping. That is a modeling choice rather than a scoring defect; the mean calibration of Equation (8) absorbs its first-order attenuation, it is shared by every arm, and it cancels to first order in comparisons.

3.3 What Downstream Inherits

Every forecast in this paper passes through Equation (11) before any QLIKE number is computed. Every downstream comparison **[pending: enumerate the downstream analyses by their final section labels (bucket attribution, block structure, model classes) once those sections stabilize]** scores its arms on raw-scale pairs produced by exactly this map, with exactly these window and residual choices.

Because the calibration pair and the second-moment coefficients are model-specific, the correction does not shift all arms by a common constant. It inflates the forecasts of an arm with larger residual variance by more, and it straightens the forecasts of a miscalibrated arm by more. Under QLIKE's asymmetric penalty, this interacts with each arm's calibration in a way that a shared correction would not. A comparison run under a different convention — an uncalibrated squared forecast, a scalar in-window smear, or no correction at all — would be a comparison of different forecast objects, and it would not necessarily induce the same ranking.

That comparison is run, restricted to exactly three conventions on the identical persisted per-bar arrays and the identical exclusion rule. Convention *none* is the naïve inverse of Section 3.1: the squared forecast with the diurnal adjustment reversed, $\widehat{RV}_t^{\text{raw}} = \hat{y}_t^2 B_t$, with no second-moment term. Convention *Duan* is the in-window scalar smear — the Duan [1983] special case identified after Equation (11) — with one scalar per evaluation window, the mean of that window's own squared fit-scale residuals, added to the squared forecast before the baseline reversal. Convention *contract* is the calibrated stack of Equations (8)–(11), the convention every headline number in this paper inherits. Table 1 reports each completed arm's pooled QLIKE under the three conventions, with within-convention ranks. The rank-stability summary is the Kendall rank correlation between the contract ranking and each alternative's: $\tau = 0.681$ against *none* and $\tau = 0.873$ against *Duan*.

Table 1: The benchmark under the three back-transform conventions. The full per-arm table is Appendix Table 5; Kendall τ of each alternative ranking against the contract ranking over all arms: none 0.681, traditional Duan 0.873.

	No correction	Traditional Duan	Contract
OLS on HAR + calendar (benchmark)	0.24099	0.23165	0.22519

The correction is treated as part of the evaluation contract: it is frozen here, before the first comparison, and no downstream section revisits it. Any future change to the second-moment model — including the learned correction contemplated for this slot — would constitute a change of contract, and would require re-scoring every comparison that inherited the old one.

4 Where the Information Lives, and What Extracts It

The reference point for this section is the per-bar least-squares fit of the HAR ladder plus calendar dummies, QLIKE 0.22519. This section measures two quantities: the change in QLIKE when each exogenous feature bucket is added to that backbone, and the overlap between the buckets' individual contributions. The buckets are defined in **[pending: cross-ref to the panel/descriptive section defining the buckets]**. The same features under a richer hypothesis class are treated in **[pending: cross-ref to the linear-vs-nonlinear section]**.

Estimator. Every arm in Table 2 is a *causally tuned elastic net* with free mixing: at every retune the tuner selects both the penalty strength (α, log-spaced grid) and the ℓ_1 share ($\rho \in \{0.25, 0.5, 0.75, 1.0\}$) on a forward validation split of the current training window, so the amount of selection is chosen by the estimator rather than fixed by the author. Each arm is refit at every bar over a rolling window of 24,000 bars (500 trading days), re-selecting its penalty every 250 solves behind a 25-bar embargo on a 125-bar validation tail; no forecast's hyperparameters ever see its own future. An unpenalized minimum-norm attribution pass over the same designs exists and is reported where its contrast is the point (Section 4.4); its per-bucket table adds nothing the penalized one does not, and the estimator class here matches the machinery the rest of the paper builds on. The only design intervention is the go-live participation rule of Section 2.2, which is a statement about data existence, not numerics.

Protocol. The attribution is computed as follows.

1. *Arms.* The incumbent arm's design matrix contains the HAR ladder columns and the calendar dummies. Each bucket arm's design matrix contains those backbone columns plus the bucket's own columns, expanded through the same moving-average ladder and availability indicators as the joint arm. The joint arm's design matrix contains

the backbone columns plus all 41 exogenous series (≈ 526 columns after the moving-average ladder and availability indicators).

2. *Fit.* Each arm is fit by the rolling estimator above and produces one out-of-sample forecast per bar.

3. *Loss.* Each arm is scored by per-bar QLIKE on the shared panel ($n = 218{,}934$ thirty-minute bars, identical rows in every cell). The reported QLIKE is the mean of the arm's per-bar losses.

4. *Increments.* Δ is the arm's QLIKE minus the incumbent's QLIKE.

5. *DM test.* For each arm, the per-bar loss differential against the incumbent is $d_t = \ell_t^{\text{arm}} - \ell_t^{\text{inc}}$. The Diebold–Mariano statistic [Diebold and Mariano, 1995] is the t-ratio of $\bar{d}$ to its Newey–West HAC standard error, with the Harvey–Leybourne–Newbold small-sample correction [Harvey et al., 1997]. Negative t favours the arm.

6. *Overlap.* The overlap ratio is

$$R = \frac{\sum_b \Delta_{\text{own}}(b)}{\Delta_{\text{own}}(\text{joint})},$$

the sum of the single-bucket within-class increments divided by the joint arm's within-class increment, both measured against the same estimator's no-exogenous baseline.

4.1 Per-Bucket Gains

Table 2 reports the causally tuned elastic net on each of the exogenous buckets, on the shared panel, against the shared incumbent.

Table 2: Bucket attribution: the causally tuned elastic net on each exogenous bucket, from the unification campaign. **Panel:** the frozen panel of Section 2, identical rows in every cell; incumbent QLIKE 0.22519 (per-bar OLS on HAR + calendar). **Estimator:** elastic net with both penalty strength and ℓ_1 share re-selected causally every 250 solves, rolling window 24,000 bars (500 trading days), refit every bar. Columns participate in a window only when their feed went live before the window starts and printed within it. Δ is against the incumbent; negative DM t favours the arm.

Bucket	QLIKE	Δ vs. incumbent	DM t
all_features (joint)	0.23116	-0.00237	-2.0
moments	0.23134	-0.00219	-2.5
market_vw	0.23361	+0.00008	+0.1
market_ew	0.23382	+0.00029	+0.4
sentiment	0.23389	+0.00036	+0.5
implied_vol	0.23401	+0.00048	+0.6
vol_demand	0.23422	+0.00069	+0.9
liquidity	0.23453	+0.00100	+0.9
(no exogenous = incumbent)	0.22519	—	—

Two rows carry the table. The moments bucket is the only single bucket that rejects equal predictive accuracy with the incumbent in its favor (-0.00219, t = -2.5); every other single bucket is statistically indistinguishable from the incumbent (market VW +0.1 through liquidity +0.9), with none individually harmful at the 5% level. And the joint arm — all exogenous columns in one fit — posts the lowest QLIKE in the table (0.23116, -2.0), below every single bucket including moments. No bucket is the story; the aggregate is. Each bucket alone carries too little signal for a 24,000-bar window to demonstrate, yet stacking them all makes the design stronger, not weaker — the signature of information spread thinly across many weak columns, which the shrinkage machinery of the remainder of this section is built to pool. How far that pooling can be pushed — and what it costs to forgo it — is measured in Sections 4.2 and 4.4.

4.2 The Same Buckets under a Competent Estimator

Table 2 fixes the estimator at the elastic net. Whether that choice flatters or handicaps the buckets is its own question: the elastic net's free mixing parameter lets the tuner choose selection, and whether selection is the right tool on this design is exactly what a family comparison decides. Each bucket design is therefore also run under a causally tuned ridge — same protocol, same grids in penalty strength, no selection available at any mixing value. Table 3 reports both families side by side, with the paired test between them.

Table 3: Each bucket design under two causally tuned penalized estimators. **Panel:** the frozen panel of Section 2, identical rows in every cell; benchmark QLIKE 0.22519. DM t is against the benchmark; negative favours the arm. The final column is the paired Diebold–Mariano statistic of the elastic net against ridge *on the same design*, positive meaning ridge is the better of the two.

	Ridge		Elastic net		Enet vs. ridge
Design	QLIKE	DM t	QLIKE	DM t	DM t
moments	0.23089	-3.6	0.23134	-2.5	+1.9
liquidity	0.23431	+0.8	0.23453	+0.9	+0.9
market (EW)	0.23354	+0.0	0.23382	+0.4	+1.0
market (VW)	0.23296	-0.9	0.23361	+0.1	+2.4
sentiment	0.23357	+0.1	0.23389	+0.5	+2.0
implied vol.	0.23362	+0.2	0.23401	+0.6	+1.5
vol. demand	0.23397	+0.7	0.23422	+0.9	+1.6
all exogenous (joint)	0.23040	-4.4	0.23116	-2.0	+1.2

Two readings follow, and they are the reason this section and the estimator comparison belong together.

First, *shrinkage without selection wins on every design*. The causally tuned ridge is the strongest estimator on the joint design (0.23040, DM -4.4), stronger than any single bucket under either family, and it attains a lower pooled loss than the elastic net in every information set. All eight paired comparisons favour ridge, and three do so at conventional significance (market VW +2.4, sentiment +2.0, moments +1.9).

Second, that ordering is not yet a statement about the two *families*, for the reason given next.

Where the elastic net's deficit comes from. Because the elastic net selects its penalty causally at every retune, each scored bar can be attributed to the penalty in force when it was forecast, and the deficit can be located rather than merely totalled. It does not spread across the sample. On the 72.1% of bars whose governing penalty was interior to the grid ($\alpha \le 10^{-3}$), the elastic net *beats* tuned ridge by 3.9×10^{-4} (DM -6.70, favouring the elastic net in all eight designs). The entire pooled deficit is contributed by the 27.9% of bars whose governing penalty sat at the *top point of the grid*, $\alpha = 10^{-2}$, where it loses by 2.5×10^{-3} (DM +6.85, in all eight designs and in 18 of 24 years). The difference between the two subsets is 2.9×10^{-3} at $z = 7.8$.

A result carried by a grid endpoint invites the explanation that the grid is too small. Under the correspondence between the elastic net's parameterization and a ridge penalty — an ℓ_2 weight of $N\alpha(1-\rho)$ on the window gram, with $N = 24{,}000$ and ρ the mixing share — the elastic net's largest expressible ℓ_2 weight is $24{,}000 \times 10^{-2} \times 0.75 = 180$, while the tuned ridge arm draws from a grid reaching 1,000 and selects that top value in 41.3% of its own retunes. The elastic net cannot express the shrinkage ridge chooses, and it loses on the bars where it has run out of grid.

That explanation is testable and it is false. Re-running every design with the penalty grid extended to $\alpha = 1$ — which reaches 6 to 18 times ridge's selected weight at every mixing value, and contains the original grid as a subset — makes the elastic net *substantially worse*, not better. On the moments design the extended-grid arm loses 2.9×10^{-3} to its own narrow-grid twin (DM +7.01) and 3.3×10^{-3} to tuned ridge (DM +6.68); on the joint design, 1.7×10^{-3} (DM +4.56) and 2.4×10^{-3} (DM +3.46). Giving the estimator more room to shrink causes it to shrink worse.

This locates the asymmetry precisely, and it is not about how much shrinkage each family can express. Both tuners pin the tops of their grids — ridge in 41.3% of retunes, the elastic net in a comparable share — because on a 125-bar validation tail neither can resolve the penalty. What differs is the consequence. For ridge, the loss is monotone and flat in $\log \alpha$ near its optimum (Section 4.5), so selecting too large a penalty costs almost nothing: over-shrinking a dense coefficient vector degrades gracefully. For the elastic net, a larger penalty is partly an ℓ_1 penalty, and over-penalizing in that direction removes columns outright. The same selection error is benign in one family and destructive in the other. Selection loses here not because the data prefer dense coefficients, but because the price of a hyperparameter mistake is asymmetric — and a short causal validation window guarantees such mistakes.

What the trajectories do and do not show. The selected mixing share is bimodal — 36.7% of retunes at pure lasso, 32.5% at the minimum 0.25, mean 0.64 — and the same shape appears in every bucket. It is tempting to read this as the estimator declining to select. Two measurements argue against that reading. Holding α fixed, the mixing share accounts for only 4.2×10^{-4} of the deficit, significant in one of eight designs and reversing sign at $\alpha = 10^{-4}$; and the selected mixing share is unrelated to the volatility regime (Spearman -0.013 against the preceding window's realized volatility, $p = 0.23$). The mixing trajectory is a symptom of a validation tail too short to identify the penalty, not the channel through which the loss is incurred.

Nor is ridge exempt from that instability: it pins an endpoint of its own grid in 58.1% of retunes, against 43.3% for the elastic net. The asymmetry this section can support is therefore narrower than a family verdict — both estimators are choosing badly on a 125-bar tail, and ridge's bad choices happen to be inexpensive while the elastic net's are not.

4.3 Overlap

In Table 2 the single-bucket increments sum to a small net *positive* (harmful) quantity — one bucket helps, six are null — while the joint arm posts the table's best loss. The buckets' combined value is therefore superadditive at this window length: no overlap ratio computed from these rows is meaningful, because the individual measurements are mostly noise around zero rather than shares of a common gain. The available overlap evidence comes from earlier, better-powered batteries: under a causally tuned ridge the sum of single-bucket gains was $R = 2.08$ times the joint gain over seven buckets, and under that battery's causally tuned LightGBM arm $R = 2.24$ — individual gains summing to roughly twice the joint gain, under both estimator classes, on the prior convention. **[pending: overlap ratio recomputed from shrunk joint fits under the paper's frozen convention — the block estimators of Sections 4.5–5 provide the joint fits]**

4.4 The Joint Bucket without Shrinkage

The attribution of Table 2 is penalized. What an *unpenalized* estimator recovers from the same stacked design is its own measurement, and the strongest version of it is minimum-norm least squares: among all coefficient vectors minimizing the training-window sum of squares, the unique one of minimal ℓ_2 norm, computed through the pseudoinverse of the window gram at the machine-precision cutoff — the $\alpha \to 0^+$ limit of ridge, defined at any design rank, with no penalty and no tuned quantity of any kind. (The cutoff is not a tuned quantity either: the gram's spectrum is gapped by more than five orders of magnitude around the retention threshold, so every tolerance across four decades reproduces the reported forecasts bitwise.)

Under that estimator the joint `all_features` arm — every exogenous column in one minimum-norm fit — has QLIKE 0.23444, statistically indistinguishable from the incumbent (+1.6). The same columns under the per-block shrinkage of Sections 4.5 and 5 produce the paper's best forecasts (0.22111 at -7.7), so what the minimum-norm joint fit measures is the cost of forgoing shrinkage, not the absence of information — and that cost is larger than everything the buckets individually contain.

The estimator choice matters here, and its history is part of the record. Under the earlier plain normal-equations solve, this arm appeared catastrophically worse than the incumbent; the deficit traced to solution components in near-null directions of the design — large mutually canceling coefficients on nearly duplicate availability-ladder columns, which a one-bar feed desynchronization on a forward-filled post-early-close ghost bar (December 24, 2020, the only such disagreement in over twenty-five thousand bars) turned into a forecast of 77.8 against a target of 0.26, whose residual the window-level correction then spread across its whole window. A second incident of the same class hit the market-aggregate buckets: in one evaluation window (December 2016 – February 2017) a pair of availability-indicator columns differing only through pre-2015 history decayed to exact collinearity as that history left the rolling window, and the contaminated solve made both aggregates look catastrophically harmful. The minimum-norm solution suppresses near-null components by construction, and the dead-session grid rule removed the ghost-bar class; under the two repairs the arms score as reported.

An independent every-bar robustness design was run on a different panel: rolling 1,000-trading-day window; HAR block exactly unpenalized via Frisch–Waugh–Lovell, with only the exogenous block penalized; scored on a fixed COVID-inclusive window from January 2018 to May 2025, $n = 74{,}934$ bars. On that panel, every bucket rejects equal predictive accuracy against HAR-only (the weakest, `vol_demand`, at $t = -6.2$), and the joint set rejects equal predictive accuracy against every single bucket, at $t = -13.0$. The bucket ranking differs across the two panels: `implied_vol` has the largest single-bucket increment on the COVID-inclusive window, `moments` in Table 2.

An earlier draft asserted that the 90% model confidence set [Hansen et al., 2011] over the feature sets is the singleton containing the joint bucket. The per-bar loss series required to compute a model confidence set were not retained for the battery, so no set was computed, and the claim is not repeated here. **[pending: battery MCS over the feature sets once per-bar losses are persisted]** Together, Tables 2 and 3 and the minimum-norm control support three statements: exactly one bucket individually beats the incumbent at the 5% level while the rest are indistinguishable from it; the joint design beats every single bucket under every penalized estimator, and under no penalty at all recovers nothing; and extracting the buckets' combined information therefore requires the shrinkage machinery the remainder of the paper develops.

4.5 The Two-Block Ridge

The estimator comparisons above generalize beyond the buckets. On the *full* design — the HAR ladder, the calendar features, and the full exogenous block of 523 columns, all under the causal tuning protocol of Table 2 so that no family carries an informed penalty choice another lacks — the loss is monotone in how densely the penalty spreads the coefficients: ridge (0.23040, DM -4.4 against the incumbent) beats the elastic net (0.23056, -2.6), which beats the lasso (0.23134, -2.0). Keeping every column and shrinking produces a lower loss than selecting a subset — and Section 4.2 located why: not a preference of the data for dense coefficients per se, but the asymmetric price of a hyperparameter mistake, benign under ℓ_2 over-shrinkage and destructive when an ℓ_1 penalty deletes live columns.

The lasso's own solution path completes the diagnosis. On the full basis it zeroes 304 of 523 exogenous columns, of which 235 — 77% — are dead or constant availability indicators: most of its selection removes columns carrying no signal, which ℓ_2 shrinkage handles equally well by making them small. Where it zeroes *live* columns, QLIKE rises — the margin ridge wins by is carried there. Nor is the signal low-dimensional in a rotated basis: truncating the design to five principal components retains 72% of the feature variance and loses essentially all of the predictive signal. Many small nonzero coefficients in the coordinate basis, no concentration in any small column subset or principal subspace: the structure this paper calls *dense but weak*. A penalty-level sweep adds the final fact: with the ridge family held fixed, QLIKE falls monotonically in α across the tested grid (0.23219 at $\alpha = 0.1$ down to 0.22909 at 10, still decreasing at the edge), and the tuned arm, drawing from a grid reaching 1000, lands below every fixed point. The penalty *level* moves the loss by more than the penalty *family* does at any fixed level, and the wide design rewards far heavier shrinkage than any textbook default.

These measurements leave two facts that a single ridge penalty cannot satisfy at once. First, penalizing the HAR-plus-calendar design alone raises QLIKE relative to plain OLS (the battery's no-exogenous control arm: ridge $t = +5.0$ against the incumbent): the HAR columns are few, well-conditioned, and individually strong, so any shrinkage on them is pure bias. Second, the wide exogenous design requires heavy shrinkage — that is the head-to-head's result. A scalar penalty applies the same α to both blocks: a value large enough for the hundreds of weak exogenous columns over-shrinks the HAR columns, and a value small enough for the HAR columns under-shrinks the exogenous columns. The resolution is a penalty that differs by block.

The section's estimator is a **two-block ridge**: one jointly solved least-squares problem with a block-diagonal Tikhonov penalty,

$$(\hat{\beta}_1, \hat{\beta}_2) = \arg\min_{\beta_1,\beta_2} \|y - X_1\beta_1 - X_2\beta_2\|_2^2 + \alpha_1\|\beta_1\|_2^2 + \alpha_2\|\beta_2\|_2^2, \qquad (\alpha_1, \alpha_2) = (1, 100).$$

Block 1 (X_1) is the HAR ladder of the target, at $\alpha_1 = 1$. Block 2 (X_2) is the same HAR averaging construction applied to *every* exogenous feature — HAR-of-`all_features`, the full wide basis — at $\alpha_2 = 100$. The two blocks are not fit separately and stacked; the coefficients are estimated in one solve. The per-block penalties reduce to a single ridge by column scaling: with $\tilde{X} = \left[X_1/\sqrt{\alpha_1} \;\; X_2/\sqrt{\alpha_2} \right]$ and $\gamma = (\sqrt{\alpha_1}\,\beta_1,\ \sqrt{\alpha_2}\,\beta_2)$,

$$\|y - X_1\beta_1 - X_2\beta_2\|_2^2 + \alpha_1\|\beta_1\|_2^2 + \alpha_2\|\beta_2\|_2^2 = \left\|y - \tilde{X}\gamma\right\|_2^2 + \|\gamma\|_2^2,$$

so the two-block problem is an ordinary ridge with unit penalty on the rescaled design, and $\hat{\beta}_j = \hat{\gamma}_j/\sqrt{\alpha_j}$ for $j = 1, 2$. **[pending: how the $(\alpha_1, \alpha_2) = (1, 100)$ pair was fixed — grid/validation provenance, and its sensitivity]**

The estimator matches the measured coefficient structure clause by clause. The signal is spread across many columns, so block 2 keeps all of its columns: nothing is selected. No individual exogenous coefficient is large, so block 2 is shrunk two orders of magnitude harder than block 1, and the fit sums many attenuated contributions. The HAR columns are few and strong, so block 1 is penalized only at $\alpha_1 = 1$; setting $\beta_2 = 0$ reduces the model to ridge at $\alpha = 1$ on the backbone, which differs from the OLS–HAR incumbent of Section 2 only by that nominal penalty. The estimator contains no selection step and no tuned dial: the penalties are fixed constants, not optimized quantities.

The two-block ridge scores QLIKE 0.22941, with Diebold–Mariano -9.0 against the OLS–HAR incumbent. Under the documented convention (penalties $1/3 \times 10^3$, 250-day window) the same structure scores 0.22775 at -9.6. These are the first scored runs of this exact specification. **[pending: DM of the two-block ridge against single-penalty ridge on `all_features` — computed from the same per-bar losses at harvest]**

The next section extends this estimator by one block. Section 5 identifies the nonlinearity the trees use, writes it as an explicit product block, and estimates the final three-block model. The added block keeps all of its columns, carries its own fixed penalty, and nothing is selected.

4.6 Time-Weighting the Window

Every estimator above weights the 24,000 bars of its training window equally. A direct measurement of coefficient stability motivates asking whether it should: on non-overlapping windows across the panel, the backbone's coefficient vector is 94–97% directionally invariant over twenty-two years, while the exogenous block's loses roughly 40% of its direction — once, on a timescale of 62–250 trading days — and then holds a plateau for two decades. There is no ongoing decay to track, but the oldest bars in a two-year window do describe a slightly different exogenous coefficient than the newest.

Exponentially down-weighting the window tests this. The gram and cross-moment are accumulated with weight w^{t-s} on bar s, truncated to the same 24,000 bars so that $w = 1$ reproduces the flat window exactly, with the half-life $H = \log 2/\log(1/w)$ the only new quantity. Three variants were run. With H *fixed* at half the window (12,000 bars), the estimator improves on the flat window: QLIKE 0.21280 against the three-block ridge's 0.21942, a paired increment of -0.00033 at DM -2.8. With H fixed at 3,000 bars it is worse (+0.00087 at +2.4), and — the now-familiar pattern — letting the causal tuner *select* H from a grid is worse still (+0.00145 at +3.7): the selected half-life flickers across the entire grid from one retune to the next, and the flicker costs several times what the right fixed value earns. Mild fixed down-weighting of the oldest half-window helps; handing the forgetting rate to a 125-bar validation tail destroys the gain. This is the same asymmetry the penalty comparison of Section 4.5 measures, appearing on a temporal dial.

One negative result belongs beside this. The hypothesis that serial correlation of the residuals explains why validation-selected penalties so far exceed in-sample risk-optimal ones was tested and fails on measurement: the out-of-sample one-step residuals are nearly white (integrated autocorrelation time 1.88, first-lag autocorrelation 0.03), because per-bar refitting leaves consecutive forecasts sharing almost no estimation error. A correction built on that hypothesis is inert by construction, and the discrepancy between in-sample and validation-selected shrinkage remains attributed to the loss geometry — the QLIKE criterion on levels against the squared-error criterion the estimator is risk-optimal for — which this paper measures but does not resolve.

One further negative result closes the linear discussion. A factor summary of the panel cannot replace the exogenous block outright: a ladder of two-block models — the backbone plus K rotated directions of the exogenous set, K from 5 to the full live rank, under both eigenvalue and frozen predictive orderings, each causally tuned — trails the matched two-block model on the raw columns by roughly 2×10^{-3} at every rung, *including full rank* (+0.00154 at DM +4.3). At full rank under a uniform block penalty a rotation cannot change a ridge fit, so the gap is not a property of the rotation: it traces to the construction having re-standardized its inputs by frozen frame-window statistics where the production path uses the rolling scaler, an identity verified directly (rotating the production-scaled columns reproduces the raw block's fitted values to 4×10^{-15}). Rolling standardization of the inputs is therefore worth on the order of 2×10^{-3} against frozen standardization — several times the size of most increments this paper reports — and it, not the choice of basis, ordering, or width, is the largest single design decision in the factor machinery.

5 Nonlinearity and the Product Block

Section 4.5 established the linear results: the exogenous signal is dense and weak, uniform heavy shrinkage is the measured linear response, and the two-block ridge is that response. This section measures what nonlinear models add. It has three parts. First, gradient-boosted trees beat the best linear arm on every feature set, by a small margin (Section 5.1). Second, decompositions of the tree fits attribute the bulk of that margin to additive-plus-pairwise structure, and the pairwise part to products of columns the panel already contains, gated on the session clock. Third, those products are written down as explicit columns, appended to the linear design, and shrunk as a block of their own (Section 5.2). The section ends with the resulting three-block ridge, the final rung of the block ladder this paper builds (two blocks in Section 4.5).

5.1 The Tree Edge

Protocol. The evidence is the causal-tuning battery of Section 4: five estimators fully crossed with the nine feature buckets on the shared panel ($n = 218{,}934$ thirty-minute bars, identical rows in every cell). The two nonlinear arms are the gradient-boosted tree ensembles

LightGBM [Ke et al., 2017] and XGBoost [Chen and Guestrin, 2016]. The linear arms are the penalized regressions of Section 4.5. Every arm, tree and linear alike, reselects its own hyperparameters on the same periodic causal schedule: each reselection uses only data available before the reselection date, so no forecast's configuration uses information from that forecast's future. The shared comparator is the per-bar OLS–HAR incumbent (Section 2.4).

Results. LightGBM has the lowest QLIKE on all nine feature sets. Ridge, the best linear arm, is second on all nine. The ranking reverses on no bucket. On the richest set, `all_features`, LightGBM scores 0.12604 and ridge scores 0.12788, a model-class gap of 0.00184; across the nine sets the tree-minus-linear gap ranges from 0.00184 to 0.00358. An exact sign test over the nine sets gives $p = 0.0039$. A bound-based pairwise test resolves all nine matched comparisons in the tree's favour after Holm adjustment; the test uses the fact that the incumbent cancels from the tree-minus-linear loss differential, so the pairwise standard error is bounded by the sum of the two arms' standard errors with no assumption on their dependence. The `all_features` pair is the weakest at $|t| \geq 2.19$, $p \leq 0.029$; the other eight range from $|t| \geq 3.73$ to $|t| \geq 5.91$. The model-class gap is roughly a quarter of the information effect of Section 4.

The function-class premium at fixed information. The battery's no-exogenous `baseline` row uses the incumbent's information set exactly: HAR lags and calendar features, nothing else. On that row the causally tuned LightGBM scores 0.13118 ($t = -5.6$) and XGBoost scores 0.13163 ($t = -5.5$). Measured against the base-2 OLS ladder at 0.13332, the premium from relaxing the linear functional form is 0.00215. Measured against the causally tuned ridge on the same features (0.13445), the premium is 0.00327. We carry the smaller figure, 0.00215, forward. On this same row the three linear arms are the only arms in the battery with higher QLIKE than the incumbent.

Dispersion of the improvement. The Diebold–Mariano statistics [Diebold and Mariano, 1995, Harvey et al., 1997] against the incumbent on `all_features` order the two arms opposite to their QLIKE levels: ridge reaches $t = -19.9$ and LightGBM reaches $t = -15.4$. The DM statistic is the mean per-bar loss improvement divided by the standard error of that improvement, so a larger $|t|$ at a smaller mean improvement implies a smaller bar-to-bar dispersion of the improvement: ridge's per-bar improvement has the smaller dispersion, the tree's the larger mean.

[decision pending]

Where the nonlinearity lives. Three decompositions of the tree fits were computed in an earlier phase of this project.

Interaction order. A functional-ANOVA decomposition of a fitted depth-eight boosted tree attributes approximately 55% of its explainable variance to additive terms, 9% to pairwise interactions, and 36% to a higher-order remainder concentrated in no identifiable interaction. Within the additive part, the HAR block accounts for 96%.

Saturation at pairwise order. On the deployed residual-stack panel of the earlier build (n = 194,934; levels not comparable to the battery), an expressivity ladder holds the base model fixed and varies only the correction stage. A purely additive correction scores 0.12757. Adding bilinear pairwise interactions scores 0.12108. Nonlinear pairwise shapes score 0.12080. The deployed bagged additive-plus-pairwise stage (a GA2M in the sense of Lou et al. 2013) scores 0.12033. An Optuna-searched unrestricted-depth XGBoost correction stage scores 0.12094, which is worse than fitting no correction at all (0.12081). The ladder improves monotonically up to pairwise order; the unrestricted stage does not improve on the pairwise stages.

Distillation into products. On an earlier build of the same panel family, a penalized linear base scores 0.12516 and a residualized boosted-tree correction improves it to 0.12414. Adding to the base a single hand-written interaction — the HAR lags multiplied by a late-day session dummy, six columns — scores 0.12457, which recovers 58% of the tree correction's improvement. The full set of HAR-by-six-regime interactions scores 0.12453, or 62%. The session dummy in the recovered term marks the late-day bars at which the change of sign in HAR's persistence coefficient is documented in Section 2.

The tree experiments therefore yield a specification statement: **the panel's recoverable nonlinearity is a sparse set of pairwise products of columns the design already contains**. The rest of this section constructs that set of products as an explicit feature block.

5.2 The Product Block

The construction enumerates candidate products, selects a subset, and appends the subset to the linear design as ordinary columns, read by the same rolling ridge machinery as every other block. The interaction study that developed the block settled four design choices: which products, reselected how often, shrunk how hard, and refit how often. Its numbers were computed on the interaction-study panel — 218,934 out-of-sample bars, 2007–2024, a 250-day rolling window, HAR reference QLIKE 0.13275 — which shares its rows with the battery panel but not its window or backbone convention. The levels below are therefore not comparable to the battery tables and are quoted only for the design decisions they settle. **[pending: confirmation that the paper's production product block is exactly this construction (candidate space, selection window, column scaling) once the block is re-run under the paper's panel convention]**

Construction. The block is built in four steps. The constants below are those frozen in the campaign executor, `src/unification.py`, the code that runs the campaign's product arms; where the interaction study's description and the executor differ, the executor's construction is the one stated here, because it is what ran.

1. *Candidate set.* The base set is the target's HAR ladder, the 41 exogenous value channels at trailing windows of 1, 32, and 512 bars, and the four session-clock columns (the open, close, and overnight indicators and the hour of day) — 133 columns on the interaction-study panel. The candidate set is every pairwise product of two base columns, including squares: 8,911 candidate products.

2. *Selection target.* A backbone-only ridge — the HAR-plus-calendar block at penalty $\alpha = 1$, refit every bar — is walked forward over the first $2W$ rows of the panel, where

W is the arm's rolling fit window: 24,000 bars (500 days) under the stated convention of Section 5.3, 12,000 bars (250 days) under the documented convention. The selection target is that ridge's out-of-sample residual on rows $[W, 2W)$.

3. *Scoring and selection.* Each base column is centered over rows $[W, 2W)$. Each candidate is scored by the absolute correlation, over those rows, between the product of its two centered parents and the selection target. The $k = 100$ candidates with the largest scores are kept. Selection runs once; the set is never reselected; and every arm carrying the block is evaluated only from row $2W$ onward, so no scored bar overlaps the selection window.

4. *Column scaling.* Each kept product is formed bar by bar as the product of its two parent columns as they stand in the panel (uncentered, carrying the causal prescaling of Section 2). The product column is then divided by its own trailing rolling standard deviation over a W-bar window, lagged one bar and requiring at least 1,000 observations, with the scale floored from below at 0.1 times the per-bar cross-column median of the product columns' rolling standard deviations — the floor bounds the inflation a low-dispersion trailing window can produce, the failure mode the scaling discipline of Section 2 guards against. Rows before the first valid scale, all inside the first window and before any scored bar, reuse the first valid value. A product with zero dispersion over its entire trailing window is set to zero on those bars. Every product value at bar t is a function of data available at bar t.

Set size. The interaction study's set-size ladder is unimodal: accuracy peaks at $k = 25$–100 of 8,911 and degrades by $k = 400$. The linear channel of Section 4.5 improved monotonically in the number of columns; the interaction channel peaks and degrades. The block uses $k = 100$.

Reselection cadence. Two arms were compared with identical machinery and identical coefficient refits, differing only in when selection runs. The static arm selects the 100 products once, on the first selection window (February 2001 through October 2003 in calendar time), and keeps them for the rest of the sample. The dynamic arm reselects the product set every month. The dynamic arm loses: at $k = 100$ it trails the static arm at DM $t = -3.21$ under the study's original scaling, and the sign is negative at every healthy specification tested. The dynamic set replaces 24–31% of its members per month. The product set is selected once, eighteen years before the end of the sample, and is never reselected.

Block penalty. Products of centered features have variance on the order of the product of their parents' variances, so the product block and the base block have columns on incommensurate scales. Under a single shared penalty the product block's measured contribution is unstable and can be negative; an early +43% headline for the block was an artifact of a design clip that was standing in for the missing block-wise regularization. Under a separate penalty, with the product block shrunk an order of magnitude harder than the linear block, the hundred frozen products add $\Delta R^2 = +0.0067$ over the full 516-column linear base —

a +35% relative improvement in residual-space accuracy, DM $t = +2.71$. Lighter product penalties eliminate the gain.

Coefficient refit cadence. The comparison holds the frozen block and the solver fixed and varies only how often the coefficients are re-estimated. At a monthly coefficient refit, the block's QLIKE increment is indistinguishable from zero (DM $t = +0.52$). At a daily refit, the identical frozen block's residual-space increment is +0.0108, and the raw-scale reconstruction gives ΔQLIKE $= -0.00093$, DM $t = +3.56$. The selected products are predominantly fast pairs built from one-bar quantities. The identity of the products is frozen; their coefficients are refit frequently.

Assembled, the product block is one hundred columns with four measured properties — sparse, frozen, heavily shrunk, frequently refit — and it converts the specification statement of Section 5.1 into a feature block the paper's estimator can carry.

5.3 The Three-Block Ridge

The model this section ends on is the two-block ridge of Section 4.5 with the product block appended as a third block. Each block carries its own ridge penalty inside one rolling solver:

$$\hat{y}_t = \underbrace{X_t^{\mathrm{HAR}}\beta_1}_{\alpha_1=1} + \underbrace{X_t^{\mathrm{exog}}\beta_2}_{\alpha_2=100} + \underbrace{X_t^{\mathrm{prod}}\beta_3}_{\alpha_3=1000}, \tag{13}$$

where block one is the target's own HAR ladder at $\alpha_1 = 1$; block two is the linear exogenous block of Section 4.5 at $\alpha_2 = 100$; and block three is the frozen product block at $\alpha_3 = 1000$. The estimator is implemented by column scaling. Each column in block b is multiplied by $\sqrt{\alpha_1/\alpha_b}$, and a single ridge with penalty α_1 is solved on the scaled design; a coefficient $\tilde{\beta}$ on a scaled column corresponds to $\beta = \sqrt{\alpha_1/\alpha_b}\,\tilde{\beta}$ on the raw column, and the shared penalty $\alpha_1\|\tilde{\beta}\|^2$ on the scaled coefficients equals the per-block penalties $\alpha_b\|\beta\|^2$ on the raw coefficients. After scaling, the three-block estimator runs in the same exact rank-one rolling least-squares path as every other linear model in this paper, refit at every bar. Setting $\beta_3 = 0$ recovers the two-block model exactly.

Each penalty restates a measured result. $\alpha_1 = 1$: Section 4.5 measured that shrinking the low-dimensional, fully informative HAR block raises loss, so the backbone is left essentially unpenalized. $\alpha_2 = 100$: the exogenous signal is dense and weak, and heavy uniform shrinkage was the measured optimum. $\alpha_3 = 1000$: the interaction study measured that the product block's contribution is positive and stable only when it is shrunk an order of magnitude harder than the linear block; the one-order-of-magnitude gap between α_2 and α_3 is the ratio the study's own ladder selected on its panel. No penalty was chosen by grid search on the evaluation loss.

The campaign executor runs the three-block arm under two conventions. The stated convention is Equation (13) at a 24,000-bar (500-day) rolling fit window: penalties $(\alpha_1, \alpha_2, \alpha_3) = (1, 100, 1000)$. The documented convention pins $(1, 3 \times 10^3, 3 \times 10^4)$ at a 12,000-bar (250-day) window. Under the stated convention the three-block ridge scores QLIKE 0.22111, with Diebold–Mariano -7.7 against the OLS–HAR incumbent; under the documented convention it scores 0.22652 at -10.6. The product-block *increment* — the paired per-bar comparison of the three-block against the two-block ridge on the same rows — is ΔQLIKE = -0.00006 at DM -0.2 under the stated convention, and -0.00123 at -7.1 under the documented one; under causal per-block tuning, -0.00055 at -1.4. **[decision pending]**

The product block does not carry all of the tree's fit: the functional-ANOVA remainder and the expressivity ladder's residual gap sit above pairwise order, and the block contains pairwise products only. What the block does carry runs in a linear estimator — one solver, three penalties, refit every bar.

6 Conclusion

[decision pending]

A Master Table of Campaign Arms

Table 4 lists every arm of the unification campaign: one frozen panel (Section 2), one scoring contract (Section 3). QLIKE is the pooled contract loss; Δ is the *paired* mean per-bar loss differential against the benchmark, and DM its Diebold–Mariano t-statistic, both computed on the row intersection of the two arms. Product- and transmission-block arms with a 24,000-bar window are scored from the first row outside their frozen selection and frame windows. **Protocol note.** Two row sets appear in this table: arms run under the original scoring protocol and arms run under the `oos_mult=2` protocol (chunks 9–99, 248,686 rows), which excludes the earliest out-of-sample period. Pooled QLIKE *levels* are not comparable across protocols — the excluded rows carry above-average loss, so `oos_mult=2` arms read mechanically lower. The Δ and DM columns are paired on common rows and are comparable throughout; level comparisons should be made only within a protocol. The gated two-block ridge and the *trail*-family arms are the `oos_mult=2` panel. **Contract note.** The scoring contract's second-moment layer was upgraded on 2026-08-11 from the window-level scalar to the causal adaptive recursion of Section 3 (Equation (10)). Rows harvested on Hoffman2 — the benchmark, the `oos_mult=2` panel, and every increment between them — are scored under the adaptive contract. Rows whose chunks survive only on the decommissioned cluster (the original-protocol block ridges and the conditional second-moment panel) retain their *legacy* scalar-contract levels; the two contracts shift QLIKE levels by roughly $5\text{–}8 \times 10^{-3}$, so legacy levels are not comparable to adaptive-contract levels, while paired Δ and DM within each contract remain valid.

Table 4: All campaign arms under the frozen panel and scoring contract.

Model	QLIKE	Δ	DM t
Benchmark			
OLS on HAR + calendar (benchmark)	0.22519	—	—
Single-bucket attribution (minimum-norm least squares)			
HAR + moments bucket	0.23089	-0.00264	-5.7

continued

Table 4 (continued)

Model	QLIKE	Δ	DM t
HAR + liquidity bucket	0.23471	+0.00118	+3.3
HAR + market bucket (equal-weighted)	0.23357	+0.00004	+0.1
HAR + market bucket (value-weighted)	0.23303	-0.00050	-1.2
HAR + sentiment bucket	0.23253	-0.00100	-5.2
HAR + implied-volatility bucket	0.23363	+0.00010	+0.4
HAR + volatility-demand bucket	0.23375	+0.00022	+1.2
HAR + all exogenous columns (joint)	0.23444	+0.00091	+1.6
Bucket designs under causally tuned ridge			
HAR + moments bucket	0.23089	-0.00264	-3.6
HAR + liquidity bucket	0.23431	+0.00078	+0.8
HAR + market bucket (equal-weighted)	0.23354	+0.00001	+0.0
HAR + market bucket (value-weighted)	0.23296	-0.00057	-0.9
HAR + sentiment bucket	0.23357	+0.00004	+0.1
HAR + implied-volatility bucket	0.23362	+0.00009	+0.2
HAR + volatility-demand bucket	0.23397	+0.00044	+0.7
HAR + all exogenous columns (joint)	0.23040	-0.00313	-4.4
Bucket designs under causally tuned elastic net (free mixing)			
HAR + moments bucket	0.23134	-0.00219	-2.5
HAR + liquidity bucket	0.23453	+0.00100	+0.9
HAR + market bucket (equal-weighted)	0.23382	+0.00029	+0.4
HAR + market bucket (value-weighted)	0.23361	+0.00008	+0.1
HAR + sentiment bucket	0.23389	+0.00036	+0.5
HAR + implied-volatility bucket	0.23401	+0.00048	+0.6
HAR + volatility-demand bucket	0.23422	+0.00069	+0.9
HAR + all exogenous columns (joint)	0.23116	-0.00237	-2.0
Penalized linear, full design			
Ridge, fixed $\alpha = 0.1$	0.23361	+0.00008	+0.1
Ridge, fixed $\alpha = 0.3$	0.23312	-0.00041	-0.8
Ridge, fixed $\alpha = 1$	0.23243	-0.00110	-2.1
Ridge, fixed $\alpha = 3$	0.23169	-0.00183	-3.6
Ridge, fixed $\alpha = 10$	0.23087	-0.00266	-5.4
Lasso, fixed $\alpha = 10^{-4}$	0.22950	-0.00402	-8.4
Ridge, causally tuned	0.23040	-0.00313	-4.4
Elastic net, causally tuned	0.23056	-0.00297	-2.6
Lasso, causally tuned	0.23134	-0.00219	-2.0
Block ridges			
Two-block ridge (stated penalties)	0.22941	-0.00412	-9.0
Three-block ridge (stated penalties)	0.22111	-0.01242	-7.7
Four-block ridge (stated penalties)	0.22102	-0.01251	-8.0
Two-block ridge (documented penalties)	0.22775	-0.00578	-9.6
Three-block ridge (documented penalties)	0.22652	-0.00701	-10.6
Four-block ridge (documented penalties)	0.22650	-0.00703	-10.5

continued

Table 4 (continued)

Model	QLIKE	Δ	DM t
Two-block ridge (per-block causal tuning)	0.22822	-0.00531	-11.7
Three-block ridge (per-block causal tuning)	0.21942	-0.01411	-12.8
Four-block ridge (per-block causal tuning)	0.21931	-0.01422	-12.9
Four-block ridge (trailing-standardized transmission)	0.22080	-0.01273	-8.5
Four-block ridge (trailing, factor levels only)	0.22085	-0.01268	-8.2
Four-block ridge (trailing, lead–lag flow only)	0.22098	-0.01255	-8.1
Four-block ridge (trailing transmission, per-block causal tuning)	0.21915	-0.01438	-13.3
Block ridges, `oos_mult=2` protocol (248,686 rows)			
Two-block ridge (gated rows, per-block causal tuning)	0.21431	-0.00330	-5.8
Three-block ridge (product block, per-block causal tuning)	0.21368	-0.00392	-5.6
Three-block ridge (volatility-gated product block)	0.21336	-0.00424	-6.2
Single-block diagnostics			
HAR + product block only (stated)	0.22403	-0.00950	-4.3
HAR + product block only (documented)	0.23078	-0.00275	-8.9
HAR + product block only (tuned)	0.22378	-0.00975	-6.0
HAR + transmission block only (stated)	0.22248	-0.01105	-9.8
HAR + transmission block only (documented)	0.23298	-0.00055	-2.2
HAR + transmission block only (tuned)	0.22249	-0.01104	-9.0
HAR + transmission block only (trailing)	0.22272	-0.01081	-9.3
Transmission-operator variants (all trailing-standardized)			
Symmetric part of the lagged cross-correlation	0.22084	-0.01269	-8.4
Undecomposed lagged cross-correlation	0.22081	-0.01272	-8.5
Causally refreshed frame	0.22100	-0.01253	-7.8
Frame width $K = 5$	0.22093	-0.01260	-8.2
Frame width $K = 10$	0.22087	-0.01266	-8.3
Frame width $K = 40$	0.22109	-0.01244	-7.8
Two-bar lag operator	0.22081	-0.01271	-8.7
Transmission base excluding cadence-heterogeneous aggregates	0.22088	-0.01265	-8.1
Conditional second-moment contract			
Benchmark, exogenous-state variance conditioning	0.22125	-0.01228	-12.0
Benchmark, backbone-only variance conditioning	0.22043	-0.01310	-15.3
Best model, exogenous-state variance conditioning	0.21613	-0.01740	-14.7

B Smear-Convention Sensitivity, All Arms

Table 5 reports every completed arm's pooled QLIKE under the three back-transform conventions of Section 3, with within-convention ranks. The rank agreement statistics quoted there (τ against the contract ranking: none 0.681, traditional Duan 0.873) are computed over this table.

Table 5: Pooled QLIKE per arm under the three back-transform conventions (within-convention rank in parentheses).

Arm
OLS on HAR + calendar (benchmark)
HAR + moments bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + liquidity bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + market EW bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + market VW bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + sentiment bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + implied vol bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + vol demand bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + all-features bucket (elastic net, causally tuned, free mixing, extended penalty grid)
Ridge, fixed $\alpha = 30$
Ridge, fixed $\alpha = 100$
Ridge, fixed $\alpha = 300$
Lasso, fixed $\alpha = 10^{-6}$
Lasso, fixed $\alpha = 10^{-5}$
Lasso, fixed $\alpha = 10^{-3}$
Lasso, fixed $\alpha = 10^{-2}$
Four-block ridge (trailing factor levels, $K = 40$, rank-shaped transmission penalty, causally tuned)
Four-block ridge (trailing factor levels, $K = 40$, rank-shaped transmission penalty over an extended exponent gri
Four-block ridge (causal trailing-demeaned factor levels, $K = 40$, rank-shaped transmission penalty over the exte
Four-block ridge (factor levels divided by their causal trailing SD, $K = 40$, rank-shaped transmission penalty ove
Three-block ridge (backbone + wide exogenous block + causal trailing-SD factor levels, $K = 40$, no product blo
Pure exogenous spiked-prior model: backbone plus broad exogenous block with a rank-40 frozen-factor covarianc
Frame factorial: exogenous frame, top-40 trailing-SD scores, no product block (causally tuned)
Frame factorial: product-base frame, top-40 frozen scores, fixed spiked prior, no product block (causally tuned)
Mechanism test: product-base frame, top-40 frozen scores, cross-block covariance deleted (split backbone/exog s
Mechanism test: product-base frame without HAR columns, top-40 frozen scores, no product block (causally tun
Mechanism test: product-base frame without session columns, top-40 frozen scores, no product block (causally t
Mechanism test: product-base values-only frame, top-40 frozen scores, no product block (causally tuned)
Mechanism test: product-base frame with within-family row permutation (cross-family correlation destroyed), to
Mechanism test: product-base frame, top-40 frozen spiked prior plus nonlinear product block (causally tuned)
Two-block ridge (backbone + causal trailing-SD factor levels, $K = 40$, causally tuned)
Two-block ridge with a FULL-RANK rotated backbone+exogenous block and causal trailing-SD factor scores (c
Two-block ridge with a full numerical-rank rotated backbone+exogenous block and causal trailing-SD factor scor
Full-rank factorial: exogenous frame, raw orthogonal rotation (causally tuned)
Full-rank factorial: exogenous frame, frozen mean/SD factor scores (causally tuned)
Full-rank factorial: exogenous frame, causal trailing-SD factor scores (causally tuned)
Power-law transfer test: exogenous frame, causal trailing-SD full-rank scores, rank-shaped penalty (causally tune
Full-rank refinement: exogenous frame, frozen scores, rank-shaped penalty (causally tuned)
Full-rank factorial: product-base frame, raw orthogonal rotation (causally tuned)
Full-rank factorial: product-base frame, frozen mean/SD factor scores (causally tuned)
Full-rank factorial: product-base frame, causal trailing-SD factor scores (causally tuned)
Full-rank factorial: complete-panel frame, raw orthogonal rotation (causally tuned)
Full-rank factorial: complete-panel frame, frozen mean/SD factor scores (causally tuned)
Three-block ridge with products restricted to (HAR/session) $\times$ exogenous-value interactions (causally tuned)
Four-block ridge with (HAR/session) $\times$ exogenous-value products plus causal trailing-SD factor levels, $K = 40$ (
Four-block ridge (trailing factor levels, $K = 40$, rank-shaped transmission penalty, causally tuned on HALF-DEC
Ridge, causally tuned on a half-decade grid
Two-block PCR (full rank), variance ordering, James-Stein shrinkage
Two-block PCR ($K = 30$), variance ordering, James-Stein shrinkage

References

Manabu Asai, Michael McAleer, and Marcelo C. Medeiros. Modelling and forecasting noisy realized volatility. *Computational Statistics & Data Analysis*, 56(1):217–230, 2012.

Jeremias Bekierman and Hans Manner. Forecasting realized variance measures using time-varying coefficient models. *International Journal of Forecasting*, 34(2):276–287, 2018.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009.

Matei Demetrescu, Vasyl Golosnoy, and Anna Titova. Bias corrections for exponentially transformed forecasts: Are they worth the effort? *International Journal of Forecasting*, 36(3):761–780, 2020.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Naihua Duan. Smearing estimate: A nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383):605–610, 1983.

Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.

David Harvey, Stephen Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Yin Lou, Rich Caruana, Johannes Gehrke, and Giles Hooker. Accurate intelligible models with pairwise interactions. In *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 623–631, 2013.

Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011.